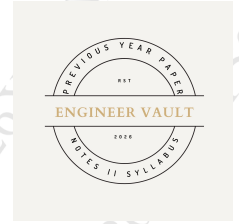


Total No. of Printed Pages: 03

SUBJECT CODE NO: - RNP-8001
FACULTY OF SCIENCE AND TECHNOLOGY
F. E. (NEP) [Group A & B] Mech, Civil, Comp. AI & ML, EEE
Examination (January/February-2025)
Applied Mathematics - I

**[Time: 3:00 Hours]****[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Q.nos 2, 3, 4, 5, 6 and 7

SECTION A**Q1 Solve the following questions: (Any 10)****20**

- a) Which of the following functions can be expressed as a Maclaurin's series?
 - a) Any differentiable function.
 - b) Functions that are differentiable at $x=0$.
 - c) Only polynomial functions.
 - d) Any continuous
- b) Taylor's theorem is valid for which types of functions?
 - a) Continuous functions
 - b) Functions that is infinitely differentiable.
 - c) Only polynomial functions.
 - d) Functions with a finite number of derivatives question/multiple
- c) When evaluating $\lim_{x \rightarrow \infty} \frac{x}{e^x}$ which indeterminate form arises?
 - a) $0/0$
 - b) ∞ / ∞
 - c) $0 \times \infty$
 - d) 1^0
- d) By the Cauchy's n^{th} root test the series $\sum_{n=1}^{\infty} u_n$ is convergent if
 - a) $\lim_{n \rightarrow \infty} (u_n)^{\frac{1}{n}} < 1$
 - b) $\lim_{n \rightarrow \infty} (u_n)^{\frac{1}{n}} > 1$
 - c) $\lim_{n \rightarrow \infty} (u_n)^{\frac{1}{n}} = 1$
 - d) 0
- e) The maximum value of the function $\sin x \cdot \sin y \cdot \sin(x+y)$ is
 - a) $\frac{3}{2}$
 - b) $\frac{\sqrt{3}}{2}$
 - c) $\frac{3\sqrt{3}}{8}$
 - d) 1

- 2

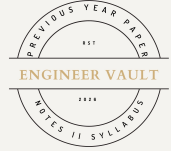
m) Find $\frac{\partial z}{\partial y}$ where $z = ax^2 + 2by^2 + 2bxy$,

a) $3by$

c) $3(ax+by)$

b) $2ax$

d) $4by+2bx$



SECTION B

- Q2** a) Use Taylors theorem to find $\sqrt{25.15}$ **05**
- b) Prove that $\lim_{x \rightarrow 0} \left(\frac{1^x + 2^x + 3^x + 4^x}{4} \right)^{\frac{1}{x}} = (24)^{1/4}$ **05**
- Q3** a) If $xu = yz, yv = zx, zw = xy$ then find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ **05**
- b) If $u = \tan^{-1} \left(\frac{x^3 + y^3}{x - y} \right)$, prove that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \sin 4u - \sin 2u$ **05**
- Q4** a) Solve: $\sin x \frac{dy}{dx} + y \cos x = 2 \sin^2 x \cdot \cos x$ **05**
- b) Solve: $\frac{dy}{dx} = \frac{\tan y - 2xy - y}{x^2 - x \tan^2 y + \sec^2 y}$ **05**
- Q5** a) Explain $\log(1+\sin x)$ in powers of x . **05**
- b) Test the convergence of $\sum_{n=1}^{\infty} \frac{1}{(n+1)^2}$ **05**
- Q6** a) If $z = x^y$ then show that $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$ **05**
- b) Divide 24 into three parts such that the continued product of the first square of the Second & cube of the third is maximum **05**
- Q7** a) $3y^2 \frac{dy}{dx} + 2xy^3 = 4xe^{-x^2}$ **05**
- b) A body of mass m , falling from rest is subjected to the force of gravity & an air resistance proportional to the square of the velocity. If it falls through a distance x & possesses a velocity v at that instant, Prove that $\frac{2kx}{m} = \log\left(\frac{a^2}{a^2 - v^2}\right)$ where $mg = ka^2$. **05**